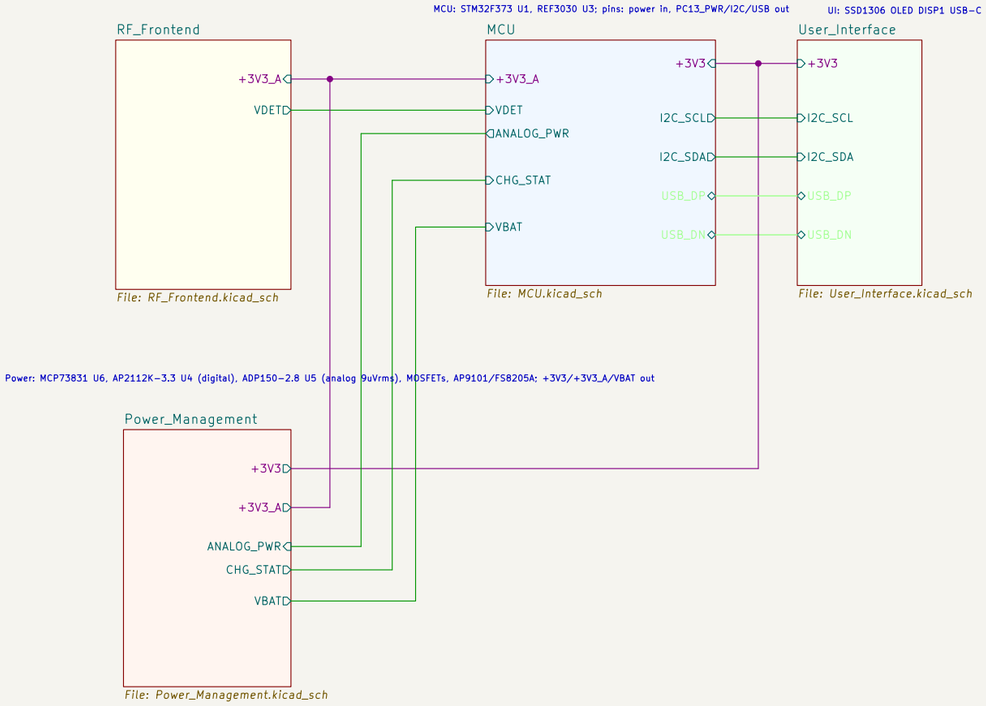
RFef to: https://www.eevblog.com/forum/rf-microwave/how-to-properly-use-these-sma-connectors/new/#new

ad8317_powermeter.kicad_sch — top-level A3: 4 hierarchical sub-sheets

RF Frontend: SMA J1, AD8317 U2, RC filter; +3V3_A in, VDET/AGND_RF out

MCU: STM32F373 U1, REF3030 U3; pins: power in, PC13_PWR/I2C/USB out

UI: SSD1306 OLED DISP1 USB-C CONN1; I2C/USB



Power: MCP73831 U6, AP2112K-3.3 U4 (digital), ADP150-2.8 U5 (analog 9uVrms), MOSFETs, AP9101/FS8205A: +3V3/+3V3_A/VBAT out

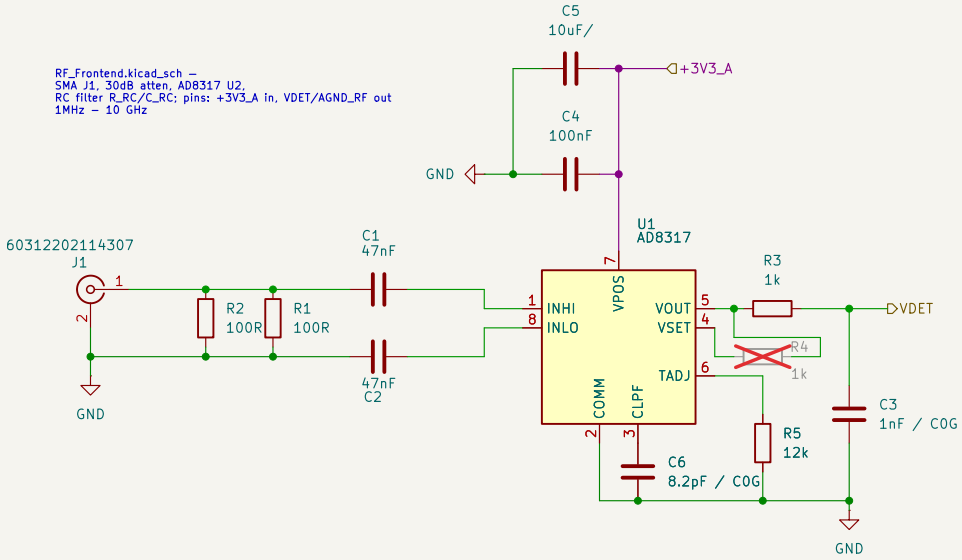
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)
Author

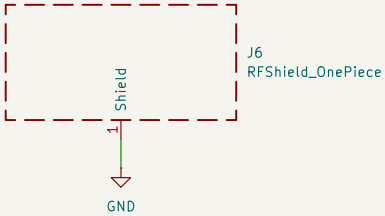
Sheet: /
File: ad8317_powermeter.kicad_sch

Title: RF Frontend

Size: A3	Date: 2026-08-07	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/5

RF_Frontend.kicad_sch —
SMA J1, 30dB atten, AD8317 U2,
RC filter R_RC/C_RC; pins: +3V3_A in, VDET/AGND_RF out
1MHz – 10 GHz





AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)

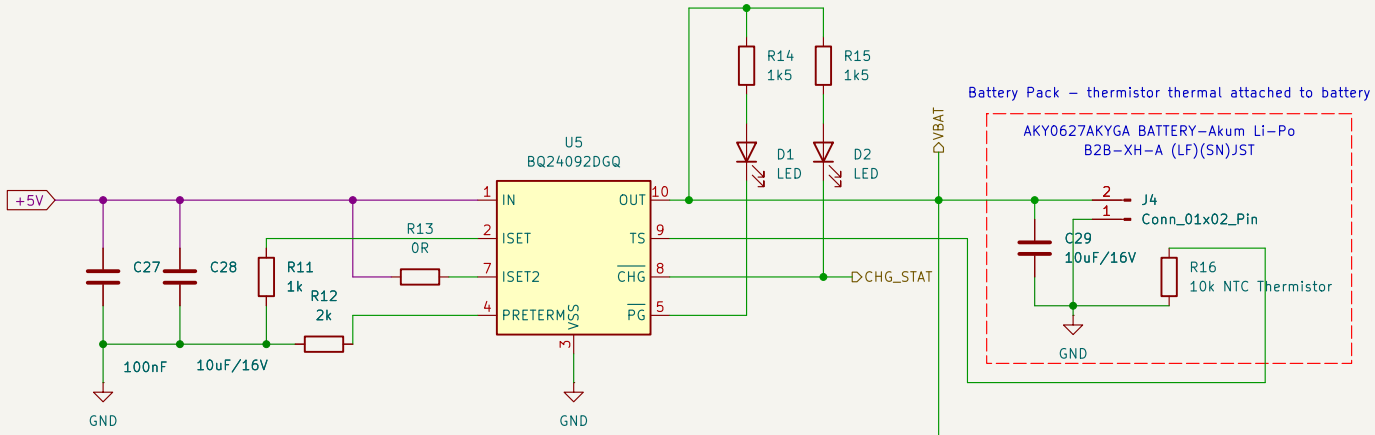
Author

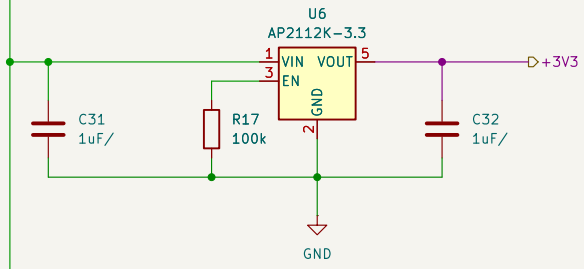
Sheet: /RF_Frontend/
File: RF_Frontend.kicad_sch

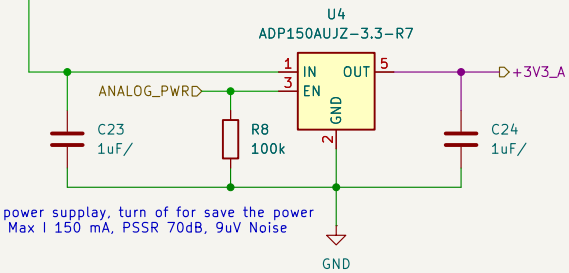
Title: RF Frontend

Size: A4	Date: 2026-07-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/5

Power_Management.kicad_sch — MCP73831 charger U6, AP2112K-3.3 digital LDO U4, ADP150AUJZ-2.8 analog LDO U5 (9uVrms), AP9101/FS8205A protection; pins: PC13_PWR in, +3V3/+3V3_A/VBAT/CHG_STAT out







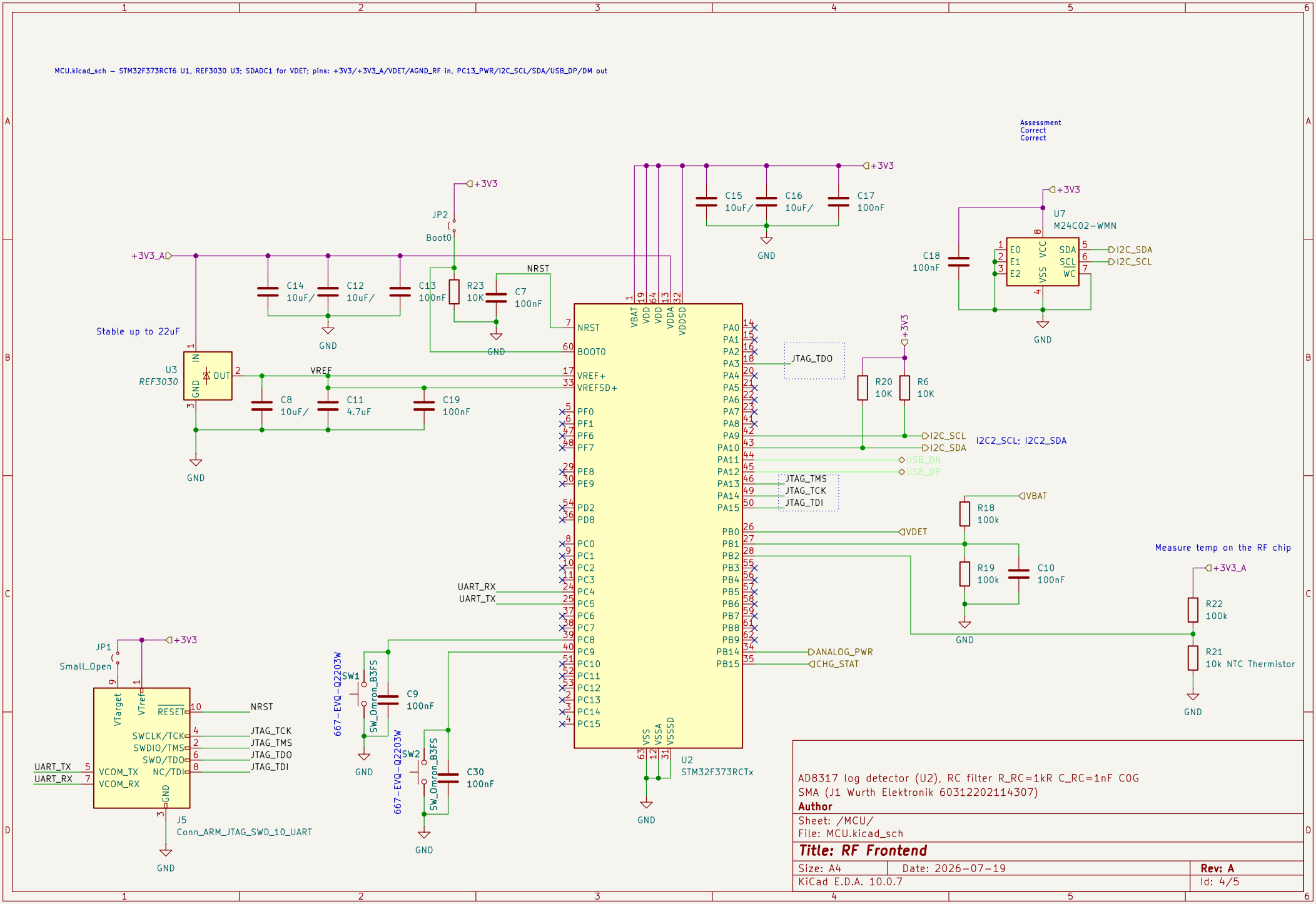
AP9101 protection (U7), FS8205A cell FETs (T_PCM), FB1-FB3 600R@100MHz
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)

Author

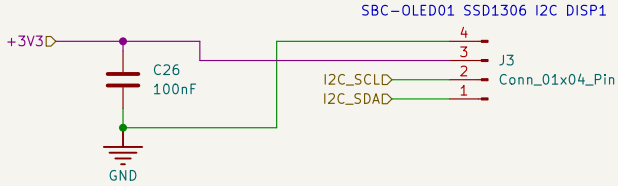
Sheet: /Power_Management/
File: Power_Management.kicad_sch

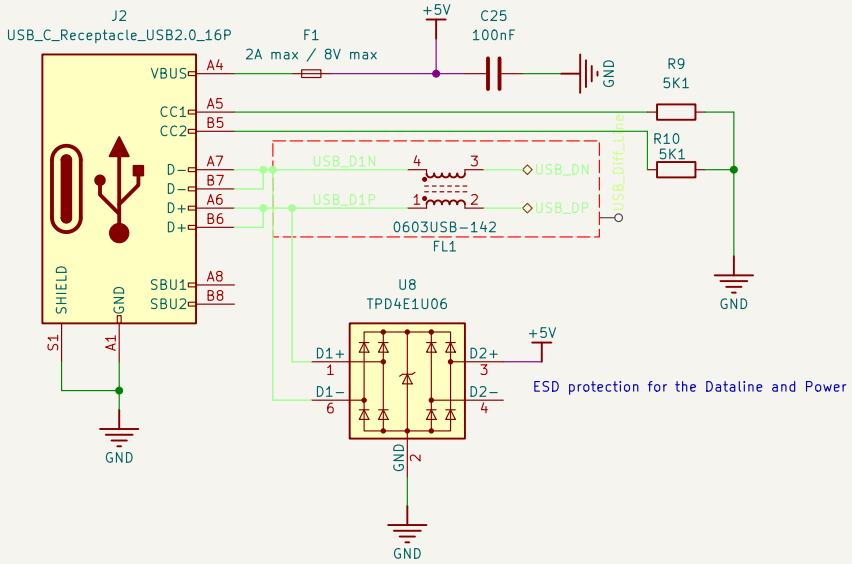
Title: RF Frontend

Size: A4	Date: 2026-07-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/5


MCU.kicad_sch — STM32F373RCT6 U1, REF3030 U3; SDADC1 for VDET; pins: +3V3/+3V3_A/VDET/AGND_RF in, PC13_PWR/I2C_SCL/SDA/USB_DP/DM out
Assessment
Correct
Correct
+3V3
+3V3_A
JP2
Boot0
C15 10uF/
C16 10uF/
C17 100nF
C14 10uF/
C12 10uF/
C13 100nF
R23 10K
C7 100nF
NRST
GND
Stable up to 22uF
U3
REF3030
IN
OUT
GND
VREF
C8 10uF/
C11 4.7uF
C19 100nF
U7
M24C02-WMN
C18 100nF
E0
E1
E2
VCC
SDA
SCL
WC
VSS
I2C_SDA
I2C_SCL
R20 10K
R6 10K
JTAG_TDO
I2C_SCL
I2C_SDA
I2C2_SCL; I2C2_SDA
USB_DN
USB_DP
JTAG_TMS
JTAG_TCK
JTAG_TDI
VBAT
R18 100k
R19 100k
C10 100nF
VDET
Measure temp on the RF chip
+3V3_A
R22 100k
R21 10k NTC Thermistor
UART_RX
UART_TX
JP1
Small_Open
VTarget
VTref
RESET
SWCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
VCOM_TX
VCOM_RX
J5
Conn_ARM_JTAG_SWD_10_UART
667-EVQ-Q2203W
SW1
SW_Omron_B3FS
C9 100nF
SW2
C30 100nF
ANALOG_PWR
CHG_STAT
U2
STM32F373RCTx
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)
Author
Sheet: /MCU/
File: MCU.kicad_sch
Title: RF Frontend
Size: A4
Date: 2026-07-19
Rev: A
KiCad E.D.A. 10.0.7
Id: 4/5

User_Interface.kicad_sch — SBC-OLED01 SSD1306 I2C DISP1, SW1 MENU PA0, SW2 NEXT PA1, USB-C CONN1 USB4110-GF-A; pins: I2C/+3V3/USB in, SW_MENU/SW_NEXT out





https://www.we-online.com/components/media/o109030v410%20ANP002d_The%20protection%20of%20USB%202.0%20applications.pdf

USB-C connector USB4110-GF-A (CONN1) — shared net with Power sheet
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)

Author

Sheet: /User_Interface/
File: User_Interface.kicad_sch

Title: RF Frontend

Size: A4	Date: 2026-07-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/5